(Following paper code and roll no. to be filled by student at the answer books)											
PAPER CODE:BAS203	ROLL NO.										

BBS COLLEGE OF ENGINEERING & TECHNOLOGY

PRAYAGRAJ
FIRST SESSIONAL EXAM (2025-2026)

COURSE: B.TECH SEM: 2nd
SUBJECT: Engg.Maths-2
MAX MARKS: 20 MAX TIME: 1:30 HR

NOTE: 1. QUESTION PAPER CONTAINS THREE SECTIONS.
2. ALL SECTIONS ARE COMPULSORY.

(SECTION A) **[5 X 1 = 5]**

Q1. ATTEMPT ALL FROM THE FOLLOWING:

a. Find the general solution of the differential equation $\frac{d^5y}{dx^5}-\frac{d^3y}{dx^3}=0$.

b. Find the Particular Integral for the following differential equation $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=e^{3x}$.

c. Find the Particular Integral for $\frac{d^2y}{dx^2}+4y=\sin 2x$.

d. Find the Laplace Transform of $f(t)=\cosh at\cos bt$.

e. Find the Laplace transform of $f(t)=3^t$.

(SECTION B) **[4 X 2 =08]**

Q2. ATTEMPT ANY FOUR OF THE FOLLOWING:

a. Find the complete solution of $\left(\frac{d^2}{dx^2}+a^2\right)y=\sec ax$.

b. Solve $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=x^2e^{-x}\cos x$.

c. Solve $\left(\frac{d}{dx}+\frac{1}{x}\right)^2 y=x^{-4}$.

d. Find the Laplace Transform of $\int_0^t e^{-t}\cos t\,dt$.

e. Find the Laplace Transform of $\frac{1-\cos t}{t}$.

f. Evaluate $L^{-1}\left(\frac{e^{-2s}}{s^2}\right)$.

(SECTION C) **[2X3.5= 7]**

Q3. ATTEMPT ANY TWO OF THE FOLLOWING:

a. Solve $t\frac{dy}{dt}+x=0$ and $t\frac{dx}{dt}+y=0$.

b. Use variation of parameter method to solve the differential equation $x^2y''+xy'-y=x^2e^x$

c. Find the Laplace transform of $f(t)=\begin{pmatrix} t-1, 1<t<2 \\ 3-t, 2<t<3 \end{pmatrix}$